

Product List Block

Summary

Product List component displays product catalogs with pricing on web pages, allowing users to view and purchase products.

References Page Url

T [Celleste Image Analysis Software | Thermo Fisher Scientific - US](#)

T [EVOS LED Light Cubes | Thermo Fisher Scientific - US](#)

T [Molecular Biology Bulk Reagents and Lab Supplies for Sample Preparation | Thermo Fisher Scientific - US](#)

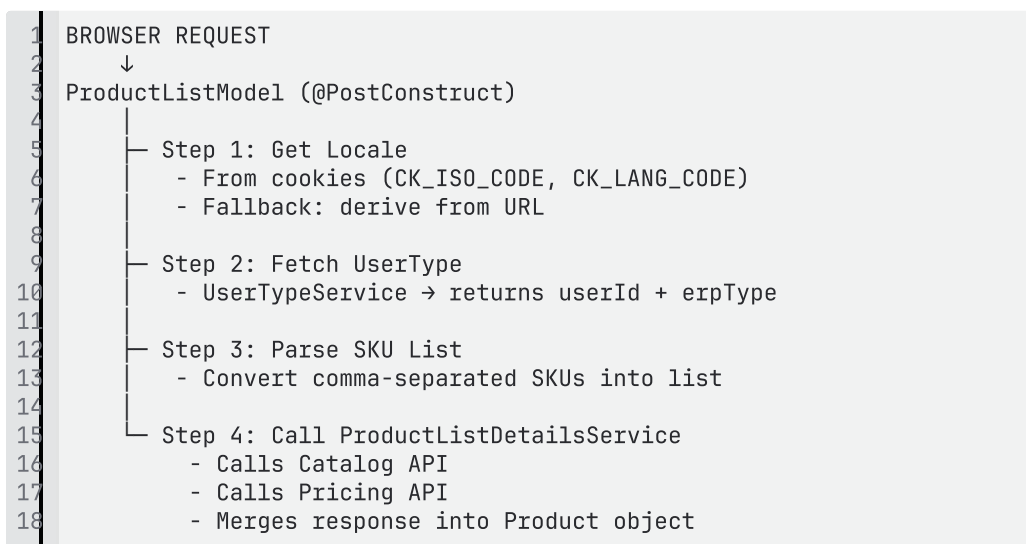
T [Transfected Cell Analysis | Thermo Fisher Scientific - US](#)

Current State

The Product List component is a flexible, data-driven AEM component that renders product information in a table format. It:

- Fetches product details via **Catalog Microservice API** based on SKU list
- Retrieves pricing via **Pricing Access API** with user-specific pricing
- Renders a dynamic HTML table with configurable columns (SKU, Name, Size, Price, Quantity)
- Supports **Add-to-Cart** via client-side JavaScript
- Handles multiple price access types (RequestQuote, OrderNow, NoPrice, etc.)

Data Flow



```
19
20     ↓
21 JSP renders product table
22     ↓
23 BROWSER DISPLAY
```

Service Layer Responsibility

`ProductListDetailsServiceImpl` performs:

- UserType retrieval
- OAuth token generation via `AuthTokenProvider`
- Catalog API call
- Pricing API call
- Consolidation into a single Product response

Frontend Responsibility

- Only renders data provided by backend
- Handles Add-to-Cart:
 - Validates quantity
 - Calls mini-cart
- No API calls from browser

EDS Approach

EDS does not support AEM-style server-side rendering. The published HTML is static.

To support dynamic product data while keeping SEO and security intact, a **two-layer architecture** is proposed:

1. Edge Worker (Presentation Layer)

- Reads product-list block from page
- Extracts SKU list and configuration
- Calls Product Microservice
- Renders HTML using returned JSON
- Injects final HTML into response

2. Product Microservice (Data Layer)

- Handles OAuth token generation
- Resolves user context (UserType)

- Calls:
 - Catalog API
 - Pricing API
 - Merges responses
 - Returns normalized JSON
-

Product List Block (Authoring in EDS)

The EDS Product List block will allow authors to configure:

- **SKU list** (comma-separated or structured input)
- **Column selection**, such as:
 - Size
 - Price
 - Quantity
 - Add-to-Cart

This ensures flexibility similar to the current AEM component while keeping authoring simple.

Important Design Decision

- Business logic **is not recommended to be implemented in Edge Worker**
- Edge should remain thin (rendering only)
- Microservice should handle all backend logic

Microservice can be implemented using:

- App Builder
 - AWS Lambda
 - Node.js service
-

Suggested Architecture View

```
1 Universal Editor
2   ↓
3 EDS Page (Product List Block)
4   ↓
5 Edge Worker
6   - Read SKUs
7   - Call Microservice
8   - Render HTML
9   ↓
10 Product Microservice
11  - Auth Token
```

```
12 - UserType
13 - Catalog API
14 - Pricing API
15 - Merge response
16   ↓
17 Backend APIs
```

Additional Notes

- Add-to-Cart remains **client-side** (same as current)
 - **TFS is also planning to introduce a microservice for fetching product details.**
 - Need to confirm:
 - Scope of that service
 - Integration approach with EDS
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Recommendation

Adopt **Edge Worker + Product Microservice pattern.**

- Keeps API logic secure and outside browser
 - Maintains SEO via server-side rendering at edge
 - Keeps edge layer simple and maintainable
 - Reuses existing backend logic effectively
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Open Questions

1. How does the **User Type API identify the user** in EDS?
 - Cookie-based / token-based / other